CS604 OPERATING SYSTEM
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1. When a process has undivided access to a shared piece of code than no other process can execute code, this state is called_____.
    - Race condition
    - Progress
    - **Mutual exclusion (page 98)**
    - Bounded waiting

1. Mutual Exclusion
If process P_i is executing in its critical section, then no other process can be executing in their critical section.

2. Critical section is a place where certain shared structure is updated. Its solution required certain precaution: one of them is the access to critical section by one process at a time only. What this condition is termed as?
    - Progress
    - Entry section
    - **Mutual exclusion (page 97)**
    - Bounded waiting

the process is in its critical section (for that shared data). The execution of critical sections must be mutually exclusive: at any time, only one process is allowed to execute in its critical section (even with multiple processors). So each process must first request

3. Which of the following is not a major advantage of multiprogramming system is
    - More than one jobs can be processed at a given time
    - CPU utilization can be increases
    - Jobs can be completed quickly
    - **If more process can exist, the rest must wait until the CPU is free and can be rescheduled (page 31)**

The objective of multiprogramming is to have some process running all the time so as to maximize CPU utilization. The objective of time-sharing is to switch the CPU among processors so frequently that users can interact with each program while it is running. A uniprocessor system can have only one running process at a given time. If more processes exist, the rest must wait until the CPU is free and can be rescheduled. Switching the CPU

4. We can suspend a foreground process by pressing_____ which sends a STOP/SUSPEND signal to the process.
    - Ctrl-w
    - Ctrl-x
    - Ctrl-y
    - Ctrl-z pg#65

Suspending a process

You can suspend a foreground process by pressing <Ctrl-Z>, which sends a STOP/SUSPEND signal to the process. The shell displays a message saying that the job

5. _____command in LINUX is used to copy file.
    - ls
    - cp pg#27
    - mv
    - mkdir

Copying Files

You can use the cp command for copying files. You can use the cp file1

6. System calls provide the interface between a_____ and the Operating System.
    - User
    - Machine
    - Process pg#15
    - Kernel

System Calls

System calls provide the interface between a process and the OS. These calls are

7. In critical section problem _____ requirement illustrates that, “If no process is executing in its critical section and some processes wish to enter their critical sections, then only those processes that are not executing in their remainder section can participate in the decision on which will enter its critical section next, and this selection cannot be postponed indefinitely. ”
    - Mutual exclusion

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- Progress pg#98
- Bounded waiting
- Slow execution

2. Progress

If no process is executing in its critical section and some processes wish to enter their critical sections, then only those processes that are not executing in their remainder section can participate in the decision on which will enter its critical section next, and this selection cannot be postponed indefinitely.

8. Batch programs are usually _____ programs.
   - Interactive
   - **Non-interactive (page 88 + Google)**
   - Foreground
   - Preemptive

are easily classified into different groups. For example, a common division is made between **foreground** (or **interactive**) processes and **background** (or **batch) processes**.

batch program A non-interactive (non-conversational) program such as a report listing or sort.

9. You can use the mv file1 file2 command to move_____.
   - **File1 to file 2 (Page 27)**
   - File 2to file 1
   - This command will not work for moving files
   - None of the option correct

Moving Files

You can use the `mv` command for moving files. You can use the `mv file1 file2` command to move file1 to file2. The following command can be used to

10. The_____ defines an operating system as a bridge between computer user and hardware for a user's convenience.
    - System-view
    - Bottom-up view
    - **Top-down view (page 03)**
    - Layered-view

There are two views about this. The top-down view is that it is a program that acts as an intermediary between a user of a computer and the computer hardware, and makes the computer system convenient to use. It is because of the operating system that users of a

11. The section of code before the critical section is called_____.
    - **Entry section (Page 97)**
    - Remainder section
    - Exit section
    - Crystal section

permission to enter its critical section. The section of code implementing this request is

called the **entry section**. The remaining code is the **remainder section**. The critical

12. _____ are used by shell commands to pass data from one shell pipeline to another, without creating temporary files.
    - Pipes
    - Message queues
    - BSD sockets
    - **FIFOs (Page 58)**

Two common uses of FIFOs are:
- In client-server applications, FIFOs are used to pass data between a server process and client processes
- Used by shell commands to pass data from one shell pipeline to another, without creating temporary files

13. _____ system call is used to create a child process.
    - **Fork (page 48)**
    - Parent
    - Child
    - None of them

end descriptor stored in pipefd[0] and write end descriptor stored in pipefd[1]. We call `fork()` to create a child process, and then use the fact that the memory image of the

14. The problem with priority scheduling algorithm is_____.
    - Deadlock
    - **Starvation (page 83)**
    - Aging
    - Nice value

A major problem with priority- scheduling algorithms is **indefinite blocking** (or **starvation**). A process that is ready to run but lacking the CPU can be considered

15. The basic purpose of_____ is to help the users to utilize the hardware resources for completing different task in a simplified manner.
    - Operating system (page 01)
    - Application software
    - All software
    - All of the given

Operating system, which manages the use of the hardware among the various application programs for the various users and provides the user a relatively simple machine to use.

16. If your processor does not have two slots empty in Pre Process file Descriptor Table, then your_____system call will fail.
    - **Pipe (page 52)**
    - Read
    - Write
    - Open

We discussed the `pipe()` system call in the notes for lecture 8. The `pipe()` system call fails for many reasons, including the following:
- At least two slots are not empty in the PPFDT—too many files or pipes are open in the process

17. Which of the following are TRUE for direct communication?
    - A communication link can be associated with N number of process(N=maximum number of processes supported by system)
    - **A communication link can be associated with exactly two processes (pg 44)**
    - Exactly N/2 links exist between each pair of processes(N=maximum number of processes supported by system)
    - Exactly two links exist between each pair of processes

A communication link in this scheme has the following properties:
- A link is established automatically between every pair of processes that want to communicate. The processes need to know only each other's identity to communicate
- A link is associated with exactly two processes.

18. POSIX is a standard develop by_____.
    - ANSI
    - **IEEE (Google)**
    - ISO
    - ACM

The **Portable Operating System Interface** (**POSIX**) is a family of standards specified by the IEEE Computer Society

19. Keeping in mind scheduling algorithm, when you go to any famous fast food franchise. On entering the store, each customer receive a number. The customer with a lowest number is served next. This algorithm is called_____.
    - Dijkstra's algorithm
    - Simple algorithm
    - Deadlock algorithm
    - **Bakery algorithm (page 102)**

The bakery algorithm is due to Leslie Lamport and is based on a scheduling algorithm commonly used in bakeries, ice-cream stores, and other locations where order must be made out of chaos. On entering the store, each customer receives a number. The customer with the lowest number is served next. Before entering its critical section, process

20. Which scheduling algorithm allocates the CPU first to the process that request the CPU first?
    - **First come First served scheduling (page 80)**
    - Shortest job scheduling
    - Priority scheduling
    - None of the given

First-Come, First-Served (FCFS) Scheduling
The process that requests the CPU first (i.e., enters the ready queue first) is allocated the CPU first. The implementation of an FCFS policy is managed with a FIFO queue. When

21. Bounded buffer is a buffer of_____ size.
    - Variable
    - **Fixed (page 41)**

not try to consume an item that has not yet been produced. The bounded buffer problem assumes a fixed buffer size, and the consumer must wait if the buffer is empty and the

22. The wait operation of the Semaphore basically works on the basic_____ system call.
    - Stop()
    - Block()
    - Hold()
    - **Wait() (page 111)**

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process from the list of the waiting processes and awakens that process. The wait operation can be defined as:

```
void wait(semaphore S) {
```

23. Priority scheduling cannot be pre-emptive.
    - True
    - **False** **(page 83)**

Priority scheduling can either be preemptive or non-preemptive. When a process

24. When a process P1 switches from the running state to the waiting state because a I/O request is being completed. This scheduling is called_____.
    - Non preemptive pg#79
    - Switching
    - Termination
    - **Preemptive**

1. When a process switches from the running state to the waiting state (for example, an I/O request is being completed)
2. When a process switches from the running state to the ready state (for example when an interrupt occurs)
3. When a process switches from the waiting state to the ready state (for example, completion of I/O)
4. When a process terminates

In 1 and 4, there is no choice in terms of scheduling; a new process must be selected for execution. There is a choice in case of 2 and 3. When scheduling takes place only under 1 and 4, we say, scheduling is **non-preemptive**; otherwise the scheduling scheme is **preemptive**. Under non-preemptive scheduling once the CPU has been allocated to a

25. The priority of a process can be changed using _____ command.
    - **Nice** **(Google)**
    - cmd
    - cat
    - grep

You can change the process priority using nice and renice utility. Nice command will launch a process with an user defined scheduling priority.

26. In_____ addressing, the recipient is not required to name the sender.
    - Symmetric

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- **Asymmetric (page 44)**
- Both(Symmetric and Asymmetric)
- None

asymmetric addressing, in which the recipient is not required to name the sender.

27. An operating system is easily portable between varying hardware designs in_____ structural approach.

- Layered
- **Micro kernels (page 19)**
- Virtual Machines
- Simple

The benefits of the micro kernel approach include the ease of extending the OS. All new services are added to user space and consequently do not require modification of the kernel. When the kernel does have to be modified, the changes tend to be fewer because the micro kernel is a smaller kernel. The resulting OS is easier to port from one hard ware design to another. It also provides more security and reliability since most services are

28. The_____ approach include the ease of extending the OS.

- Monolithic Kernel
- **Micro Kernel (page 19)**
- Macro Kernels
- Small Kernel

The benefits of the micro kernel approach include the ease of extending the OS. All

29. Which of the following information is not displayed with top command?

- **Number of threads in a process (page 64)**
- CPU Usage
- Top processes
- Nice value

`top` – Display CPU usage of processes

`top` displays information about the top processes (as many as can fit on the terminal or around 20 by default) on the system and periodically updates this information. Raw CPU percentage is used to rank the processes. A sample run of the `top` command is shown in Figure 10.6. The output of the command also shows the current time, how long the system has been up and running, number of processes running on the system and their status, number of CPUs in the system and their usage, amount of main memory in the system and its usage, and the size of swap space and its usage. The output also shows a lot of information about each process, including process ID, owner's login name, priority, nice value, and size. Information about processes is updated periodically. See the manual

30. _____processes or thread often need access to shared data and shared resources.

- Parallel
- **Concurrent (page 95)**
- Single
- Temporary

Concurrent processes or threads often need access to shared data and shared resources. If

31. To terminate a process <ctrl-c> is pressed, which signal it actually passes to process for termination?

- SIGTERM
- **SIGINT (page 66)**
- SIGKILL
- SIGTRAP

You can terminate a foreground process by pressing `<Ctrl-C>`. Recall that this key press sends the SIGINT signal to the process and the default action is termination of the process. Of course, if your foreground process intercepts SIGINT and ignores it, you

32. UNIX System V scheduling uses queues, which run _____algorithm.

- First Come First Serve
- **Round Robin (Page 90)**
- Shortest Job First
- Shortest Remaining Time First

UNIX System V scheduling algorithm is essentially a multilevel feedback priority queues algorithm with round robin within each queue, the quantum being equal to1 second. The

33. Which part of the computer system helps in managing the file and memory management system?

- **Operating system (page 12 & 13)**
- Device Drivers
- Application Software
- Hardware

The OS is responsible for the following activities in connection with memory management:

The OS is responsible for the following activities with respect to file management:

34. CPU bound processes are scheduled before short or I/O bound processes in _____ scheduling, therefore, resulting in less CPU and device utilization.

- **First Come First Serve (Page 81)**
- Round Robin
- Shortest Job First
- Shortest Remaining Time First

When FCFS scheduling algorithm is used, the **convoy effect** occurs when short processes wait behind a long process to use the CPU and enter the ready queue in a convoy after completing their I/O. This results in lower CPU and device utilization than might be possible if shorter processes were allowed to go first.

35. Performance measures of scheduling algorithms are calculated with _____ using Gantt charts, in order to evaluate an algorithm for a particular workload.

- Simulation
- Analytic evaluation
- **Deterministic modeling (Page 92)**
- Queuing Models

Deterministic modeling is a type of analytic evaluation. This method takes a particular predetermined workload and defines the performance of each algorithm for workload in terms of numbers for parameters such as average wait time, average turnaround time, and average response time. Gantt charts are used to show executions of processes. We have

36. In multi-threaded process thread () take two argument, they are used to take _____and_____.

- Program counter value, address space limit.
- **New thread ID, process name**
- Register count, program counter
- New thread function name, new thread ID **(Page 68)**

`thread()` to create a thread. This function takes two arguments: the name of a function for which a thread has to be created and a variable in which the ID of the newly created thread is to be stored. The important point to note here is that multiple threads of control

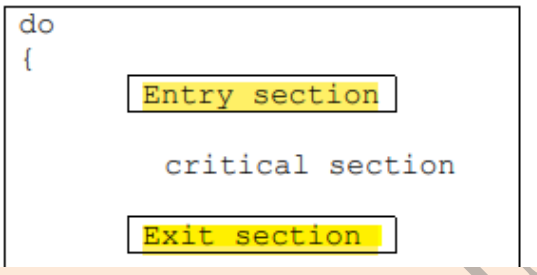
37. In the Bakery algorithm, processes are prioritized based on highest ticks among computing processes.

- True
- **False (Page 102)**

made out of chaos. On entering the store, each customer receives a number. The customer with the lowest number is served next. Before entering its critical section, process receives a ticket number. Holder of the smallest ticket number enters its critical section.

38. The solution type where in critical section we use “entry section and “exit section” is called kind of_____.
    - Hardware solution
    - Assumption
    - **Software Solution (Page 98)**
    - Operating System Solution

We discuss the software solutions first. Regardless of the type of solution, the structure of the solution should be as follows. The Entry and Exist sections comprise solution for the problem.



39. How many modes are supported in operating system?
    - 1
    - **2 (Page 8 + Google)**
    - 3
    - 4

an operating system code, we need two separate modes of operation: user mode and monitor mode (also called supervisor mode, system mode, or privileged mode). A bit,

How many modes are supported in operating system?

There are two **modes** of operation in the **operating system** to make sure it works correctly. These are user **mode** and kernel **mode**.

40. _____scheduler selects the process from the job pool and put them in main memory.
    - **Long term (Page 33)**
    - Short term
    - Medium term
    - Swapper

some fashion. The appropriate scheduler carries out this selection process. The **Long-term scheduler** (or job scheduler) selects which processes should be brought into the ready queue, from the job pool that is the list of all jobs in the system. The **Short-term**

41. A parent process calling_____ system call will be suspended until children processes terminate.

- **Wait (Page 36)**
- Fork
- Exit
- Exec

processes are able to communicate and then go their separate ways. The parent can then create more children, or if it has nothing else to do while the child runs, it can issue a `wait` system call to move itself off the ready queue until the termination of the child. The parent waits for the child process to terminate, and then it resumes from the call to

42. A time-sharing system is

- Multi user
- Multitasking
- Interactive
- **All of these (Page 06)**

A **time-sharing system** is multi-user, multi-process, and interactive system. This means

43. Pipes simply used on the command line to connect the standard input of one process to the standard input of another. Which of the following syntax is correct use of command line Linux/UNIX pipes?

- Cmd1 % cmd2 %.....%cmdN
- **Cmd1 | cmd2 |……| cmdN (Page 53)**
- Cmd1 $ cmd2 $.....$ cmdN
- Cmd1 & cmd &……&cmdN

Pipes can also be used on the command line to connect the standard input of one process to the standard input of another. This is done by using the pipe operator which is | and the syntax is as follows:

```
cmd1 | cmd2 | ... | cmdN
```

44. Which is not basic computing hardware?

- CPU
- Memory
- Compact Disc (Page 1)
- I/O Devices

1. **Hardware**, which provides basic computing resources (CPU, memory, I/O devices).

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45. Consider a system of N processes (Po, P1……..Pn-1). Each process in its critical section and process may be changing common, updating a table. No other process is allowed to execute in its section. This problem is called _____.

- Bakery algorithm
- **N-Process Critical Section (Page 102)**
- N-Mutual exclusion
- Deadlock algorithm

N-Process Critical Section Problem

In this section we extend the critical section problem of two processes to include n processes. Consider a system of n processes (P_o, P_1 …… P_{n-1}). Each process has a segment of code called a critical section in which the process may be changing common variables, updating a table, writing a file and so on. The important feature of the system in that, when one process enters its critical section, no other process is allowed to execute in its critical section. Thus the execution of critical sections by the processes is mutually

46. _____Scheduling can be made into a processor sharing approach.
    - First Come First serve
    - Round Robin (Page 86)
    - Shortest remaining time
    - Shortest Job First

policy. If the time quantum is very small, the RR approach is called the **processor sharing** and appears to the users as though each of *n* processes has its own processor

47. Linux uses_____ directory to store system configuration files.
    - /bin
    - /dev
    - /boot
    - /etc (page 23)

/etc Linux uses this directory to store system configuration files. Most files in

48. A thread shares its resources (like data section, code section, open files, signal) with_____
    - Other process similar to the one that the thread belongs to
    - Other threads that belong to similar processes
    - Other thread that belong to the same processes pg#69+Google
    - All of the mentioned

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Resource sharing: By default, threads share the memory and the resources of the process to which they belong. Code sharing allows an application to have several different threads of activity all within the same address space.

49. _____ is a piece of code in a cooperating process in which the process may update shared data (variable, file, database, etc).
    - Critical Analysis
    - Critical Section pg#97
    - Critical Path
    - Critical Code

Critical Section: A piece of code in a cooperating process in which the process may updates shared data (variable, file, database, etc.).

50. A program in execution is called a _____
    - Command
    - Process pg#12
    - Software
    - Compiler

■ Process management

A process can be thought of as a program in execution. It needs certain resources,

51. This layered approach is taken to its logical conclusion in the concept of a _____
    - Virtual Machine pg#19
    - Logical Machine
    - Physical Machine
    - None of the given

itself. This layered approach is taken to its logical conclusion in the concept of a **virtual machine** (VM). The VM operating system for IBM systems is the best example of VM

52. CPU bound processes are scheduled before short or I/O bound processes in _____scheduling therefore, resulting in less CPU and device utilization.
    - Shortest Job First
    - Shortest Remaining Time first
    - First Come First Serve (page 80)
    - Round-Robin

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When FCFS scheduling algorithm is used, the **convoy effect** occurs when short processes wait behind a long process to use the CPU and enter the ready queue in a convoy after completing their I/O. This results in lower CPU and device utilization than might be possible if shorter processes were allowed to go first.

53. Using _____ system, we can create a new process in Linux.
    - Fork pg#36
    - Exec
    - Wait
    - Exit

In order to consider these different implementations let us consider the UNIX operating system. In UNIX its process identifier identifies a process, which is a unique integer. A new process is created by the `fork` system call. The new process consists of a

54. A shell command mkfifo can be used to create a/an _____
    - Named pipe pg57
    - Message queue
    - Pipe
    - Unnamed pipe

Named pipes are created via the `mknod()` system call or `mkfifo()` C library call or by the `mkfifo` command. Here is the synopsis of the mknod() system call.

55. While using the read /write system call which data type is used to return the size of file to buffer from file descriptor fd?
    - Pipefd
    - SSIZE-MaX pg#45
    - FIFO
    - Ssize_t

`read()` attempts to read up to count bytes from file descriptor fd into the buffer starting at buf. If count is zero, `read()` returns zero and has no other results. If count is greater than SSIZE_MAX, the result is unspecified. On success, `read()` returns the

56. For reading input, which of the following system call is used?
    - write
    - rd
    - read pg#45
    - change

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processes. The read system call is used to read data from a file descriptor.

57. _____ multi-threading model provides full concurancy.

- One-to-Many
- Many-to-Many
- One-to-One pg#71
- Many-to-One

One-to-One: In this model, there is a kernel thread for every user thread, as shown in Figure 13.2. Thus, this model provides true concurrency. This means that a process

58. A solution to the critical section must satisfy the following requirements except

- Progress
- Mutual Exclusion
- Bounded Waiting
- Race Condition pg#98

Solution to the Critical Section Problem

A solution to the critical section problem must satisfy the following three requirements:

1. **Mutual Exclusion**
   If process P_i is executing in its critical section, then no other process can be executing in their critical section.
2. **Progress**
   If no process is executing in its critical section and some processes wish to enter their critical sections, then only those processes that are not executing in their remainder section can participate in the decision on which will enter its critical section next, and this selection cannot be postponed indefinitely.
3. **Bounded Waiting**

59. Which is not a parameter of pthread_create().

- Function
- Return status pg#73
- Thread ID
- Thread attributes

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```
int pthread_create(pthread_t *threadp, const pthread_attr_t *attr,
                   void* (*routine)(void *), arg *arg);
```

where, 'threadp' contains thread ID (TID) of the thread created by the call, 'attr' is used to modify the thread attributes (stack size, stack address, detached, joinable, priority, etc.), 'routine' is the thread function, and 'arg' is any argument we want to pass to the thread function. The argument does not have to be a simple native type; it can be a 'struct' of whatever we want to pass in.

60. Preemptive SJF (Shortest Job First) scheduling is sometimes called _____ scheduling.
    - shortest time-last
    - shortest remaining-time-first pg#82
    - smallest time-first
    - biggest time-first

finish its CPU burst. Preemptive SJF scheduling is sometimes called **shortest-remaining-time-first** scheduling.

61. The file descriptor for Standard Input (stdin) is _____
    - 0 pg#52
    - 1
    - 2
    - 3

d files are attached to the terminal on which a process runs. The descriptors for standard files are known as **standard file descriptors**. Standard files, their descriptors, and their default attachments are:

- Standard input: 0 (keyboard)

62. Rather than maximizing CPU utilization and use of peripheral devices, _____ systems are for maximizing user convenience and responsiveness.
    - Single-user pg#4
    - Multi-user
    - Batch
    - Time-sharing

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Single-user systems

A computer system that allows only one user to use the computer at a given time is known as a **single-user system**. The goals of such systems are maximizing user convenience and responsiveness, instead of maximizing the utilization of the CPU and peripheral devices. Single-user systems use I/O devices such as keyboards, mice, display

63. The Operating system is a layer of software between _____ and _____.
    - Hardware, software application Google
    - Kernel, hardware
    - Dos, Windows
    - Windows, Kernel

The operating system is an integral **layer** in the modern computer. **The operating system** (**OS**) is specialized computer **software** that allocates memory and manages **system** resources. When a computer is turned on, **the OS** is loaded into memory and works as an abstraction **layer between** the physical hardware and the **software**.

64. You can have a thread wait for another thread with the same process by using the ____ system call.
    - pthread_join() pg#73
    - pthread_exit()
    - pthread_create()
    - pthread_terminate()

Joining a Thread

You can have a thread wait for another thread within the same process by using the pthread_join() call. Here is the syntax of this call.

65. In indirect communication between processes P and Q_____
    - There is another process R to handle and pass on the messages between P and Q.
    - There is another machine between the two processes to help communication
    - There is mailbox to help communication between P and Q pg#44
    - A link is associated with exactly two processes

Indirect Communication

With indirect communication, messages can be sent to and received from mailboxes. Here, two processes can communicate only if they share a mailbox. The send and receive

66. Which process can be affected by other processes executing in the system?
    - cooperating process pg#41+Google

- child process
- parent process
- init process

The concurrent processes executing in the operating system may be either independent processes or cooperating processes. A process is independent if it cannot affect or be affected by any other process executing in the system. Clearly any process that shares

Explanation: A cooperating **process can be affected by other processes executing in the system**. Also it **can affect other processes executing in the system**. A **process** shares data with **other processes**, such a **process** is known as a cooperating **process**.

67. _____ are used to show executions of processes.
    - Gantt charts pg#85
    - Time slice
    - Quantum
    - Sequence

respectively. The first Gantt chart shows execution of processes according to the longest-

68. The ready queue in RR (Round Robin) scheduling algorithm is referred to as _____.
    - Time Quantum
    - Time Slice
    - LIFO (Last in First Out)
    - Circular Queue pg#85

The **round-robin (RR) scheduling algorithm** is designed especially for time-sharing systems. It is similar to FCFS scheduling but preemption is added to switch between processes. A small unit of time, called a **time quantum** (or **time slice**) is defined. The ready queue is treated as a circular queue. The CPU scheduler goes around the ready queue, allocating the CPU to each process for a time interval of up to 1 time quantum.

69. Linux redirection features can be used to _____and attach other files with them for a single execution of a command.
    - detach the default files from stdin and stdout
    - detach the default files from stdin and stderr
    - detach the default files from stdin, stdout and stderr pg#55
    - detach the default files from stdout, and stderr

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Input, output and error redirection in UNIX/Linux

Linux redirection features can be used to detach the default files from stdin, stdout, and stderr and attach other files with them for a single execution of a command. The act of

70. The scheduling of _____ are done by the operating system.
    - Kernel threads (Page 70)
    - User level threads
    - Both Kernel and User level threads
    - None of the above

Kernel threads are supported directly by the operating system. The kernel performs the scheduling, creation, and management in kernel space; the kernel level threads are hence slower to create and manage, compared to user level threads. However since the

Kernel-level **threads are scheduled by the OS**, and each **thread** can be granted its own

71. A _____ is an integer variable that, apart from initialization is accessible only through standard atomic operations.
    - Mutual Exclusion
    - Semaphore pg#108
    - Mutex
    - Busy waiting

semaphore. A **semaphore** S is an integer variable that, apart from initialization is accessible only through two standard atomic operations: wait and signal. These

72. _____ displays information about the top processes.
    - ls
    - cs
    - top pg#64
    - cd

top – Display CPU usage of processes

top displays information about the top processes (as many as can fit on the terminal or

73. In critical section problem, each process must first request permission to enter its critical section. The section of code implementing this request is called the _____ .
    - Remainder section

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- Entry section pg97
- Exit section
- Sub section

in its critical section (even with multiple processors). So each process must first request permission to enter its critical section. The section of code implementing this request is

called the **entry section**. The remaining code is the **remainder section**. The critical

74. _____ scheduling allows a process to move between queues.
    - Round Robin
    - First Come First Serve
    - Multilevel Feedback Queue pg#89
    - Shortest Remaining Time First

Multilevel Feedback Queue Scheduling

Multilevel feedback queue scheduling allows a process to move between queues. The

75. To terminate a process <ctrl-c> is pressed, which signal it actually pass to process for termination?
    - SIGTERM
    - SIGINT pg#66
    - SIGKILL
    - SIGTRAP

Terminating a process

You can terminate a foreground process by pressing `<Ctrl-C>`. Recall that this key press sends the SIGINT signal to the process and the default action is termination of the

76. The time it takes for the dispatcher to stop one process and start another running is known as the _____ .
    - Dispatch latency pg#79
    - Dispatch time
    - Stop time
    - Sleep time

The time it takes for the dispatcher to stop one process and start another running is known as the **dispatch latency**.

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77. In _____ inter process communication, a sender mention the name of a recipient.

- Indirect
- Direct pg#43
- Automatic
- Synchronous

Direct Communication

With direct communication, each process that wants to communicate must explicitly name the recipient or sender of the communication. The send and receive primitives are

78. Consider three processes in scheduling. Here are the waiting time for three processes P1=5,P2=10,P3=3.Which one of the following is correct average waiting time per process?

- 3
- 2
- 18
- **6 Conceptual**

Here are the waiting times for the three processes and the average waiting time per process.

Waiting times **P1 = 5; P2 = 10; P3 = 3**

Average waiting time: **(5+10+3)/3 = 6**

Average waiting time= 6

79. The priorities of processes in the _____ group remain fixed.

- Kernel pg#90
- User

The priorities of processes in the Kernel Group remain fixed whereas the priorities of

80. _____ algorithm is used for solving n-process critical section problem.

- Bankers
- Bakery pg#102
- Babbles
- None of the given

executions of critical sections. Each process must request permission to enter its critical section. Many solutions are available in the literature to solve the N-process critical section problem. We will discuss a simple and elegant solution, known as the Bakery algorithm.

81. The situation in which no context switching is required in multiprocessor system is referred to as_____.
    - Busy waiting
    - Spin lock pg#110
    - Interrupt
    - Scheduler

may be able to use productively. This type of semaphore is also called a **spinlock** (because the process spins while waiting for the lock). Spinlocks are useful in multiprocessor systems. The advantage of a spinlock is that no context switch is required when a process must wait on a lock, and a context switch may take considerable time.

82. _____system call is used to write to a file or FIFO or any other IPC channel.
    - Read
    - Write pg#45
    - Open
    - Fork

The `write()` system call is used to write to a file. Its synopsis is as follows:

```
#include <unistd.h>
ssize_t write(int fd, const void *buf, size_t count);
```

83. In LINUX/UNIX environment Ali want to know the number of processes running on the system and their status, number of CPUs in the system and their usage, amount of main memory and its usage. Which of the following command will help in this regard?
    - $ ps
    - $ gcc
    - $ fifo
    - $ top pg#64

`top` – Display CPU usage of processes

`top` displays information about the top processes (as many as can fit on the terminal or around 20 by default) on the system and periodically updates this information. Raw CPU percentage is used to rank the processes. A sample run of the `top` command is shown in Figure 10.6. The output of the command also shows the current time, how long the system has been up and running, number of processes running on the system and their status, number of CPUs in the system and their usage, amount of main memory in the system and its usage, and the size of swap space and its usage. The output also shows a

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page for the top command for more information by using

```
$ top
```

84. IN the result displayed by ls –l command.5th column shows
    - Group of the owner
    - File size in bytes pg#26
    - Time last updated
    - Owner of the file

The output of the ls -l command gives you the following information about a file:

- 1st character: type of a file
- Rest of letters in the 1st field: access privileges on the file
- 2nd field: number of hard links to the file
- 3rd field: owner of the file
- 4th field: Group of the owner
- 5th field: File size in bytes

85. In Unix/Linux by default the standard input file is attached to the
    - Mouse
    - Keyboard pg#52
    - Light pen
    - Joystick

standard files: standard input, standard output, and standard error. By default, standard files are attached to the terminal on which a process runs. The descriptors for standard files are known as **standard file descriptors**. Standard files, their descriptors, and their default attachments are:

- Standard input: 0 (keyboard)

86. In which of the following system multiple user are allowed to used the computer simultaneously?
    - Single user
    - Two user
    - Three user
    - Multi user pg#6

Time-sharing systems

A **time-sharing system** is multi-user, multi-process, and interactive system. This means that it allows multiple users to use the computer simultaneously. A user can run one or

87. In the layered approach of operating system_____.
    - Bottom Layer(0) is the user interface
    - Highest layer(N)is the user interface pg#18

- Bottom Layer(N) is the hardware
- Highest Layer(N) is the hardwar

top of lower layer. The bottom layer is the hardware; the highest layer (layer N) is the user interface. A typical OS layer (layer-M) consists of data structures and a set of

88. A heavy weight process_____.
    - Has multiple threads of execution
    - Has a single thread of execution pg#67
    - Multiple or single thread
    - None of the mentioned

operating system resources, such as open files and system. A traditional (heavy weight) process has a single thread of control. If a process has multiple threads of control, it can

89. Is a solution of the problem of indefinite blockage of low-priority processes.
    - Starvation
    - Deadlock
    - Aging pg#84
    - None of these

Aging is solution to the problem of indefinite blockage of low-priority processes. It

90. The child process can
    - Be a duplicate of the parent process pg#36
    - Never be a duplicate of the parent process
    - Cannot have another program loaded into it
    - Never have another program loaded into it

The child process is a duplicate of the parent process.

91. FIFO's(also known as named pipes)are used for communication between_____ on UNIX/Linux system.
    - Unrelated processes
    - Related processes pg#49
    - Related and Unrelated
    - Parent-child

Unix/Linux IPC Tools

The UNIX and Linux operating systems provide many tools for interprocess communication (IPC). The three most commonly used tools are:

- **Pipe:** Pipes are used for communication between related processes on a system, as

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92. _____ Command is used to change the directory.

- Is
- Cp
- Cd pg#219
- Mv
  - Change Directory — cd

93. In Unix/Linux environment, Ayesha wants to know the complete picture of current processes in her session. Which of the following command will help her in this regard?

- $ ps (Page 63)
- $ gcc
- $ top
- $ fifo

ps gives a snapshot of the current processes. Without options, ps prints information

94. Which of the following conditions must be satisfied to solve the critical section problem?

- Mutual Exclusion
- Progress
- Bounded waiting
- All of the mentioned pg#98

Solution to the Critical Section Problem

A solution to the critical section problem must satisfy the following three requirements:

1. **Mutual Exclusion**
   If process P_i is executing in its critical section, then no other process can be executing in their critical section.
2. **Progress**
   If no process is executing in its critical section and some processes wish to enter their critical sections, then only those processes that are not executing in their remainder section can participate in the decision on which will enter its critical section next, and this selection cannot be postponed indefinitely.
3. **Bounded Waiting**
   There exists a bound on the number of times that other processes are allowed to enter their critical sections after a process has made a request to enter its critical section and before that request is granted.

95. The _____ process creates two FIFOs, FIFO2, and opens FIFO1 for reading and FIFO2 writing

- Server pg#60
- Client
- Both server and client
- None of the given options

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We call these processes client and server. The server process creates two FIFOs, FIFO1 and FIFO2, and opens FIFO1 for reading and FIFO2 for writing. The client process opens

96. Use of semaphore create problem of busy waiting , this wastes CPU cycle that some other process may be able to use productively. This type of semaphore is also called _____.
    - semaphore S
    - spinlock pg#109
    - locking semaphore
    - mutex

shared among many processes. Busy waiting wastes CPU cycles that some other process may be able to use productively. This type of semaphore is also called a **spinlock** (because the process spins while waiting for the lock). Spinlocks are useful in

97. Kernel is responsible for scheduling the user level threads
    - True
    - False pg#70

management with no support from the kernel. Since the kernel is unaware of user-level threads, all thread creation and scheduling are done in the user space without the need for

98. Which of the following of two operations are provided by the IPC facility?
    - Write & Delete message
    - Delete & Receive message
    - Send & Delete message
    - Receive & send message Google

Explanation: **Two operations provided by the IPC facility** are receive and send messages. Exchange of data takes place in cooperating **processes**.

99. Access of variable semaphore is possible only through two atomic operation _____ and _____
    - TestAndSet, Swap
    - Lock, Key
    - Boolean, integer
    - Wait, signal pg#108

semaphore. A **semaphore** S is an integer variable that, apart from initialization is accessible only through two standard atomic operations: wait and signal. These operations were originally termed P (for wait) and V (for signal). The classical definitions

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100. Kill command is used to terminate a process. The syntax of the command is kill {-signal}PID when executed without a signal number, the command sends the_____ signal to the process.

- SIGINT
- SIGKILL
- SIGTERM pg#66
- None of the given option

number 2 (which is also called SIGINT) to the process with ID 1234. The default action for a signal is termination of the process identified in the command line. When executed without a signal number, the command sends the SIGTERM signal to the process. A

101. Given below to statement can be categorized in some sort of message passing technique. This type is named as_____

1.Send (A message)

2.Receive (B message)

- Synchronization
- Direct communication pg#43
- Explicit buffering
- Send by copy communication

Direct Communication

With direct communication, each process that wants to communicate must explicitly name the recipient or sender of the communication. The send and receive primitives are defined as:

- Send(P, message) – send a message to process P
- Receive(Q, message) – receive a message from process Q.

102. Cooperating processes never share any data, code, memory or state.

- True
- **False (Page 41)**

affected by any other process executing in the system. Clearly any process that shares data with other processes is a cooperating process. The advantages of cooperating

The **cooperating processes can** cooperate with each other using **shared data** such as memory, variables, files, databases etc. Critical section is used **to** provide **data** integrity and writing is mutually exclusive **to** prevent inconsistent **data.** 11-Oct-2018

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103. _____ Command display the status of the process.

- Is
- Ps pg#63
- Gcc
- Cat

ps – Display status of processes

ps gives a snapshot of the current processes. Without options, ps prints information about processes owned by the user. Some of the commonly used options are -u, -e, and

104. A process having multiple threads of control implies_____.

- It can do more than one task at a time pg#67
- It can do only one task at a time but much faster
- It has to use one thread per process
- None of the above

operating system resources, such as open files and system. A traditional (heavy weight) process has a single thread of control. If a process has multiple threads of control, it can do more than one task at a time. Figure 12.1 shows processes with single and multiple

105. For undivided and uninterrupted testing and setting of semaphore, uni-processor systems tend to_____.

- **Disable interrupts (Page 109)**
- Use bakery algorithm
- Use spinlock
- Use signal

In a uni-processor environment, to ensure atomic execution, while executing wait and signal, interrupts can be disabled. In case of a multi-processor environment, to ensure

106. A _____ enables a user process to request the operating system to execute a privileged instruction for it.

- Interrupt
- Library call
- System Call
- **Trap (page 8)**

such as communicating with I/O devices. If a user process tries to execute a privileged instruction, a trap should be generated and process should be terminated prematurely. At the same time, a piece of operating system code should be allowed to execute privileged instructions. In order for the CPU to be able to differentiate between a user process and

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107. We can terminate a thread explicitly by either returning from the thread function or by using the _____ call.

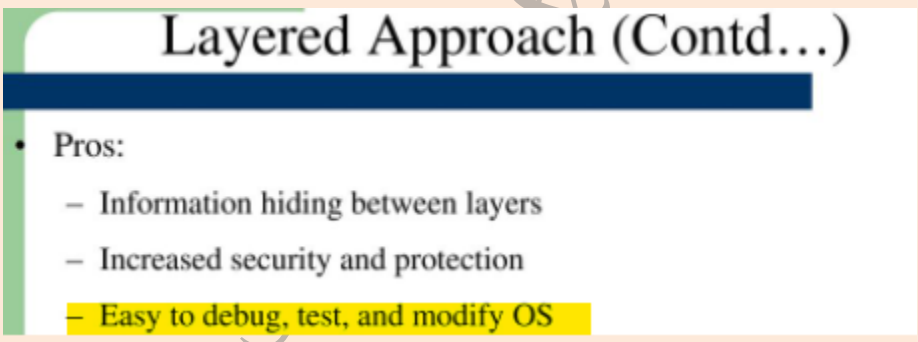
- express()
- pthread_exit() (page 73)
- win()
- slow()

You can terminate a thread explicitly by either returning from the thread function or by using the `pthread_exit()` call. Here is the syntax of the `pthread_exit()` call.

108. Ali is an operating system designer. One user requirement regarding OS is easy to debug and modify. In your opinion which one of the following OS structure Ali needs to follow?

- Simple OS Approach
- Micro kernels
- **Layered approach (Page 18 +Google)**
- Virtual machine

The main advantage of the layered approach is modularity. The layers are selected such that each uses functions and services of only lower layers. This approach simplifies debugging and system verification.



109. _____ Is not system call in Linux.

- Read
- Open
- **Mkfifo (Page 57)**
- Mknod

UNIX/Linux manual pages for the `mknod()` system call, the `mkfifo()` library call, and the `mkfifo`, `ps`, and `top` commands

110. _____ is also called swapper .

- Long term
- Short term

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- **Medium term pg#34**
- Swap space

Some operating systems such as time-sharing systems may introduce a **medium-term scheduler**, which removes processes from memory (and from active contention for the CPU) and thus reduces the degree of multiprogramming. At some later time the process can be reintroduced at some later stage, this scheme is called **swapping.** The process is swapped out, and is later swapped in by the medium term scheduler. Swapping may be necessary to improve the job mix, or because a change is memory requirements has over committed available memory, requiring memory to be freed up. As shown in Figure 5.7, the work carried out by the swapper to move a process from the main memory to disk is known as swap out and moving it back into the main memory is called swap in. The area on the disk where swapped out processes are stored is called the **swap space.**

111. After fork() system call is made, parent and child process have their separate copy of_____.
    - Environment
    - Nice value
    - File mode creation mask
    - **File descriptors pg#38**

After the `fork()` system call the parent and the child share the following:
- Environment
- Open file descriptor table
- Signal handling settings
- Nice value
- Current working directory
- Root directory
- File mode creation mask (umask)

112. Critical section problem is to _____ the concurrent execution of cooperating process.
    - Update
    - **Serialize (Page 97)**
    - Stop
    - Start

Critical Section Problem: Serialize executions of critical sections in cooperating processes.

113. Round Robin algorithm is most suitable for _____ .
    - **Time sharing system (Page 85)**
    - Real time system and batch systems
    - Running batch progress

- Expert system

The **round-robin (RR) scheduling algorithm** is designed especially for time-sharing systems. It is similar to FCFS scheduling but preemption is added to switch between

114. Concurrent processes must be synchronized to prevent _____ .

- Mutual exclusion
- **Race condition (Page 96)**
- Critical Section
- Waiting

A situation like this, where several processes access and manipulate the same data concurrently and the outcome of the manipulation depends on the particular order in which the access takes place, is called a **race condition**. To guard against such race conditions, we require synchronization of processes.

- To prevent race conditions, concurrent processes must be **synchronized**.

115. In instruction TestAndSet mutual exclusion implementation is done by declaring a Boolean variable lock_____

- Initialized as zero
- Initialized as true
- Initialized as 1
- **Initialized to false (Page 106)**

If the machine supports TestAndSet instruction, then we can implement mutual exclusion by declaring a Boolean variable lock, initialized to false. The structure of

116. The kernel is a computer program that manages_____ request from software

- **Input/output (Google)**
- Process
- Hardware
- Software

In computing, **the kernel is a computer program that manages** input/output **requests from software**, and translates them into data processing instructions for the central

117. _____is used in real time operating systems.

- Non-preemptive scheduling
- **Preemptive scheduling (Google)**
- Dispatching scheduling

- FCFS scheduling

Preemptive scheduling is the most commonly **used scheduling** algorithm in **real-time systems**. Here, the tasks are prioritized and the task with the highest priority among all

118. Which of the following provide interface to access the services of the operating system?
- API
- **System call (PAGE 15+Google)**
- Library call
- Assembly Instructions

System calls provide the interface between a process and the OS. These calls are

Explanation: The **system** call **provides** an **interface** for user programs to **access the services of the operating system** through the API (Application Program **Interface**).

119. In process management some of the jobs can be suspended for some time in order to give other job a chance to be in execution. Which command can be used to place back a suspended command?
- **fg (Page 65)**
- top
- ps
- bg

You can use the `fg` command to resume the execution of a suspended job in the foreground or move a background job into the foreground. Here is the syntax of the

120. ~symbol in the pathname ~/docs/cs604/handouts represented _____.
- Root directory
- **Login directory pg#22**
- Parent directory
- Child directory

When you log in, the system places you in a directory called your **home directory** (also called **login directory**). You can refer to your home directory by using the ~ or $PATH in Bash, Bourne shell, and Korn shells and by using $path in the C and TC shells. Shells also understand both **relative** and **absolute pathnames**. An absolute pathname

121. Which of the following statement is not true regarding the cooperating processes?
- It provides an environment to allow concurrent user to access same resources.

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- It provide an environment to run parallel processes to speedup computation.
- It construct the system in a modular fashion, dividing the system functions into separate processes or threads.
- **It may affect or be affected by any other process executing in the system. (Page 41)**

processes or cooperating processes. A process is independent if it cannot affect or be affected by any other process executing in the system. Clearly any process that shares

122. A process 'A' that has finished working but its parent process has also finished its execution. In this state the process 'A' will be called as _____process.

- Child
- Thread
- **Zombie (Page 39 +Google)**
- Fork

A **zombie process** is a process that has terminated but whose exit status has not yet been received by its parent process or by init. Sample code showing the use of `fork()` and

Zombie state: When a process is created in UNIX using fork() system call, the address space of the Parent process is replicated itself. If the parent process calls wait() system call, then the execution of the parent is suspended until the child is terminated. in this scenario the state of the child process is known as the Zombie state.

This Zombie state occurs because of the child processes, where the entry is still needed to allow the parent process to read its child's exit status: once the exit status is read via the wait system call, the zombie's entry is removed from the process table and this Zombie state said to be "reaped".

123. The nice value helps in assigning ____ to a process.

- **Priority Google**
- Weight
- Time
- Scheduling

Nice is a command in Unix and Linux **operating systems** that allows for the adjustment of the "Niceness" **value** of processes. Adjusting the "niceness" **value** of processes allows for setting an advised CPU priority that the kernel's scheduler will use to determine which processes get more or less CPU time. 09-Sep-2013

124. In critical section problem _____ requirement illustrates that, "There exists a bound on the number of times that other processes are allowed their critical sections after a process has made a request to enter its critical section and before that request is granted."

- Progress

- Mutual Exclusion
- **Bounded Waiting pg#98**
- Race Condition

3. Bounded Waiting

There exists a bound on the number of times that other processes are allowed to enter their critical sections after a process has made a request to enter its critical section and before that request is granted.

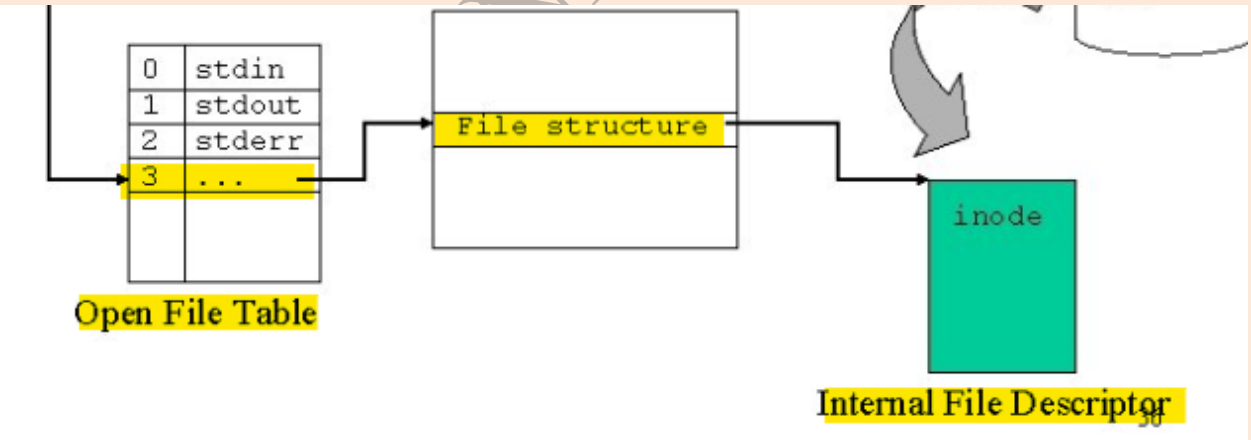
125. When process opens its first file explicitly it will get descriptor number _____ .

- 1
- 2
- **3 (Google)**
- 4

the first file descriptor that you explicitly `open` will be 3, so you'll get away with it!

- Standard input: 0 (keyboard)
- Standard output: 1 (display screen)
- Standard error: 2 (display screen)

Descriptor 3 is used for opening the file because 0,1 ,2 are standard descriptors



126. _____is a solution to critical section problem.

- Banker's algorithm
- **Lamport's bakery algorithm (Google)**
- Safety algorithm
- Dijikstra's algorithm

general case of N process. **Bakery Algorithm** is a **critical section solution** for N processes. The **algorithm**



127. Which of the following is used to show the end of the file in UNIX?

- Ctrl+A
- **Ctrl+D (Google)**
- Ctrl+O
- Ctrl+E

the "**end-of-file**" (**EOF**) **key** combination can be used to quickly log out of any terminal. CTRL-D is also used in **programs** such as "at" to signal that you have finished typing your commands (the **EOF** command).

Ctrl D tells the terminal that it should register a EOF on standard input, which bash interprets as a desire to exit. **Ctrl** + **D** (^**D**) means end of file.

128. _____is used to request the OS by the process to take an I/O or initiating child process.

- **System call (Page 195)**
- Interrupt
- Trap
- Signal

In a UNIX system, mmap() system call can be used to request the operating system to memory map an opened file. The following code snippets show "normal" way of doing file I/O and file I/O with memory mapped files.

all processes in UNIX are created by the fork system call. The initiating process is termed the parent and the newly generated process the child.

129. _____ is a segment of code that access a shared resource like data structure or device that must not be concurrently accessed by more than one thread of execution.

- Multithreading
- Context switching

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- Critical section (Page 97 +Google)
- Pipelining

When a process executes code that manipulates shared data (or resource), we say that the process is in its critical section (for that shared data). The execution of critical sections must be mutually exclusive: at any time, only one process is allowed to execute in its critical section (even with multiple processors). So each process must first request

A **critical section** is a piece of code that accesses a shared resource (data structure or device) that must not be concurrently accessed by more than one thread of execution.

130. Under_____ once the CPU has been allocated to a process the process keeps the CPU until either it switches to the waiting state, finishes its CPU burst, or terminates.

- **Non-preemptive scheduling pg#79**
- Preemptive scheduling
- Force scheduling
- Dynamic scheduling

is **preemptive**. Under non-preemptive scheduling once the CPU has been allocated to a process the process keeps the CPU until either it switches to the waiting state, finishes its CPU burst, or terminates. This scheduling method does not require any special hardware

131. The round-robin RR scheduling algorithm is designed especially for_____.

- **Time-sharing system pg#85**
- Batch system
- Single-user systems
- Nice systems

Round-Robin Scheduling

The **round-robin (RR) scheduling algorithm** is designed especially for time-sharing systems. It is similar to FCFS scheduling but preemption is added to switch between

132. In producer-consumer problem synchronization is required. On which shared area this synchronization actually effect?

- Counter
- **Buffer pg#94**
- Entry section
- Exit section

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are used. In the producer-consumer problem that was discussed earlier, the version only allows one item less than the buffer size to be stored, to provide a solution for the buffer to use its entire capacity of N items is not simple. The producer and consumer share data structure 'buffer' and use other variables shown below:

133. A process is_____ if it cannot affect or be affected by any other process executing in the system.

- **Independent pg#41**
- Dependent
- Both(independent and dependent)
- None of the given

processes or cooperating processes. A process is independent if it cannot affect or be affected by any other process executing in the system. Clearly any process that shares

134. Command-line interpreter is also called_____ in some operating system.

- Kernel
- **Shell pg#13+ Google**
- Signal
- API

logs on (on time sharing systems). This program is sometimes called the **command-line interpreter** and is often known as the **shell**. Its function is simple: to get the next

135. When processes are generated using fork () system call and then after they are coordinated with each other using IPC channel. They are utilizing a separate address space for each process (parent. child) .Kernel resources and IPC channel. This makes it a more heavy. Which strategy can be used to make it light?

- Termination of child process when these become more
- Scheduling of each process
- **Use of threads (Page 67)**
- **Use** of batch systems

1. The `fork()` system call is expensive (it requires memory to memory copy of the executable image of the calling process and allocation of kernel resources to the child process)
2. An inter-process communication channel (IPC) is required to pass information between a parent process and its children processes.

These problems can be overcome by using threads.

136. All UNIX and LINUX systems have one thing in common which is_____.

- Set of system calls (Google)

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- Set of commands
- Set of instructions
- Set of text editors

Explanation: As we know that, **all UNIX and LINUX systems have one thing in common**; they use the **same set of system calls**. .. Explanation: Many commands and **system calls**

137. First_____ entries in pre-process file descriptor table are used as soon as the process is created.
- 1
- 2
- 3 (Page 52)
- 4

Three files are automatically opened by the kernel for every process for the process to read its input from and send its output and error messages to. These files are called **standard files**: standard input, standard output, and standard error. By default, standar d files are attached to the terminal on which a process runs. The descriptors for standard files are known as **standard file descriptors**. Standard files, their descriptors, and their default attachments are:

- Standard input: 0 (keyboard)
- Standard output: 1 (display screen)
- Standard error: 2 (display screen)

138. _____ directory includes essential system boot files including the kernel image.
- /bin
- **/boot (Page 23)**
- /dev
- /etc

/boot This directory includes essential system boot files including the kernel image .

139. A_____ signal is generated when a write performed to fifo that no process is opened for reading.
- SIGFIFO
- **SIGPIPE (Page 58)**
- Named PIPE
- Unnamed PIPE

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Unlike a pipe, a FIFO must be opened before using it for communication. A write to a FIFO that no process has opened for reading results in a SIGPIPE signal. When the last

140. Consider a scenario of CPU protection,_____ is added to the operating system in order to detect and avoid loop in a user program.

- I/O
- **Timer (Page 11)**
- Base and limit register
- Turning interrupt enable off

maintains control. We must prevent the user program from getting stuck in an infinite loop or not calling system services and never returning control to the CPU. To accomplish this we can use a **timer**, which interrupts the CPU after specified period to ensure that the operating system maintains control. The timer period may be variable or

141. The process id returned to the child process after successful fork system call execution is_____.

- **0 (Page 37)**
- 1
- 2
- 3

process to communicate easily with the child process. On success, both processes continue execution at the instruction after the `fork` call, with one difference, the return code for the `fork` system call is zero for the child process, while the process identifier

142. Semaphore S is a/an_____ type of variable to use as synchronization tool.

- Float
- Double
- **Integer (Page 108)**
- Boolean

complex problems. To overcome this difficulty we can use a synchronization tool called a semaphore. A **semaphore** S is an integer variable that, apart from initialization is

143. In the bakery algorithm to solve the critical section problem_____.

- Each process is put into a queue and picked up in an ordered manner
- **Each process receives a number(may or may not be unique) and the one with the lowest number is served next (Page 102)**
- Each process gets a unique number and the one with the highest number is served next

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- Each process gets a unique number and the one with the lowest number is served next

Unfortunately, the bakery algorithm cannot guarantee that two processes (customers) will not receive the same number. In the case of a tie, the process with the lowest ID is served first. If processes Pi and Pj receive the same number, if i < j, then Pi is served first; else

144. The BSD sockets are used for communication between related or unrelated processes on the same system or_____ on different systems.

- **Unrelated processes (Page 50)**
- Related processes
- Both(related and unrelated)
- Sibling process

BSD Socket: The BSD sockets are used for communication between related or unrelated processes on the same system or unrelated processes on different systems,

145. While a process executes its critical section, other entering processes loop in a continual function in their entry sections, thus causing_____.

- Bounded waiting
- **Busy waiting (Page 109)**
- Race condition
- Mutual exclusion

requires **busy waiting**. While a process is in its critical section, any other process that tries to enter its critical section must loop continuously in the entry code. This continual

146. A major problem with priority-scheduling algorithm is _____.

- **Starvation (Page 83)**
- Conclusion
- Aging
- Termination

A major problem with priority- scheduling algorithms is **indefinite blocking** (or **starvation**). A process that is ready to run but lacking the CPU can be considered

147. The manual pages can be read in Linux using_____ command.

- **Man (Page 24)**
- wan
- desc
- help

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software updates occur. **/usr/man** is where the manual pages are kept.

148. The creating process is called a_____ process while the new process are called the _____of the process.

- Children, parent
- **Parent, children (Page 35)**
- Zombie, single
- None of the given

course of its execution. The creating process is called a **parent process** while the new processes are called the **children** of that process. Each of these new processes may in

149. UNIX system V Scheduling algorithm in every second, the priority number of all those processes that are in the main memory and ready to run is updated by using the following formula:

- Priority # = Threshold priority + nice
- Priority # = (Recent CPU Usage)/2 + nice
- Priority # = (Recent CPU Usage)/2 + Threshold priority
- **Priority # = (Recent CPU Usage)/2 + Threshold priority + nice (Page 90)**

Every second, the priority number of all those processes that are in the main memory and ready to run is updated by using the following formula:

Priority # = (Recent CPU Usage)/2 + Threshold Priority + nice

150. An operating system is a /an _____ that manages the execution of user programs to prevent errors and improper use of a computer.

- Disk
- Application program
- **Control program (Page 03)**
- Physical unit

various I/O devices and programs. An operating system is a control program that manages the execution of user programs to prevent errors and improper use of a computer.

151. State of process transits from running to ready because of _____.

- I/O
- **Interrupt (Page 79)**
- Fork()
- Scheduler dispatch

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When a process switches from the running state to the ready state (for example when an interrupt occurs)

152. Taking the CPU from one process and giving the CPU to another process is termed as

- **Context Switching (Page 31)**
- Dispatching
- Swapping
- Tracking

exist, the rest must wait until the CPU is free and can be rescheduled. Switching the CPU from one process to another requires saving of the context of the current process and loading the state of the new process, as shown in Figure 5.4. This is called **context switching**.

153. In priority-scheduling algorithm equal priority processes are scheduled in order.

- **FCFS (First Come First Serve) (Page 83)**
- LIFO (Last in First Out)
- SJF (Shorted Job First)
- SRTF (Shortest Remaining Time First)

priority (smallest integer ≡ highest priority). Equal priority processes are scheduled in FCFS order. The SJF algorithm is simply a priority algorithm where the priority (p) is the

154. A/an _____ is software generated call caused by error, or when a user request for an operating system service.

- **Trap (Page 7)**
- Interrupt
- Signal
- System call

A **trap** (or an *exception)* is a software-generated interrupt caused either by an error (division by zero or invalid memory access) or by a user request for an operating system service.

155. In process synchronization, if both the producer and consumer attempt to update the buffer concurrently, the machine language statements may get interleaved. Interleaving depends upon how the procedure and consumer processes are _____.

- Separated
- **Scheduled (Page 96)**
- Stopped
- Interrupted

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If both the producer and consumer attempt to update the buffer concurrently, the machine language statements may get interleaved. Interleaving depends upon how the producer and consumer processes are scheduled. Assume counter is initially 5. One

156. Critical section problem can be solved by using how many ways?

- 4
- **3 (Page 97)**
- 2
- 1

There can be three kinds of solution to the critical section problem:

- Software based solutions
- Hardware based solutions
- Operating system based solution

157. The write() system call may not fail for _____ reason(s).

- Invalid argument
- File size limit for process or for system would exceed
- Disk is full
- **Valid argument (Page 51)**

The `write()` system call

The call may fail for various reasons, including the following:

- Invalid argument
- File size limit for process or for system would exceed
- Disk is full

158. If a parent exits terminating its children processes, these children are handed over to _____ process.

- **Init (Page 37)**
- Root
- Daemon
- Sibling

child, so that the parent can tell which of its possibly many children has terminated. If the parent terminates however all its children have assigned as their new parent, the `init` process. Thus the children still have a parent to collect their status and execution

159. The_____ system call suspends the calling process until one of the immediate children terminate.

- **Wait (Page 39)**

- Signal
- Trap
- Interrupt

The wait system call suspends the calling process until one of the immediate children terminate, or until a child that is being traced stops because it has hit an event of interest.

160. The correct command when pipes are used on the command line to connect the standard input of one process to the standard input of another is _____.

- Cmd1 % cmd2% ……. %cmdN
- **Cmd1 | cmd2 | ……. | cmdN (Page 53)**
- Cmd1 $ cmd2 $ ……. $ cmdN
- Cmd1 & cmd2 & ……. & cmdN

Pipes can also be used on the command line to connect the standard input of one process to the standard input of another. This is done by using the pipe operator which is | and the syntax is as follows:

```
cmd1 | cmd2 | ... | cmdN
```

161. A _____ system has well defined, fixed time constraints, and if the system does not produce output for an input within the time constraints, the system fail.

- Operating
- **Real-time (Page 7)**
- Limited
- Android

A **real time system** has well defined, fixed time constraints, and if the system does not produce output for an input within the time constraints, the system will fail. For instance, it would not do for a robot arm to be instructed to halt after it had smashed into

162. In round-robin (RR) scheduling algorithm the CPU scheduler goes around the ready queue, allocating the CPU to each process for time interval of up to ___ time quantum.

- 4
- 3
- 2
- **1 (Page 85)**

ready queue is treated as a circular queue. The CPU scheduler goes around the ready queue, allocating the CPU to each process for a time interval of up to 1 time quantum.

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163. In hardware multiprocessor environment two instructions are executed _____ which are swap and TestAndSet.

- **Atomically (Page 105)**
- Automatically
- Manually
- By force

We can use these special instructions to solve the critical section problem. These instructions are TestAndSet (also known as TestAndSetLock; TSL) and Swap. The semantics of the TestAndSet instruction are as follows:

```
boolean TestAndSet(Boolean &target)
{
  boolean rv=target;
  target=true;
  return rv;
}
```

The semantics simply say that the instruction saves the current value of 'target', set it to true, and returns the saved value.

The important characteristic is that this instruction is executed atomically. Thus if two

164. We can perform the solution to critical section problem by allowing only one process to enter at a time but another solution hold some instruction use. One of them is TSL, how it is written in program?

- **TestAndSet (Boolean &target) (Page 105)**
- Testandset(Bolean &target)
- TSL (Bolean &target)
- TSL ()

```
boolean TestAndSet(Boolean &target)
```

165. The segment of code in which the process may change common variables, update tables, write into files is known as _____

- Program
- **Critical section (Page 102)**
- Non-critical section
- Synchronizing

processes. Consider a system of n processes (P_o, P_1 ..... P_{n-1}). Each process has a segment of code called a critical section in which the process may be changing common variables, updating a table, writing a file and so on. The important feature of the system

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166. When sender never block because it has an infinite length storage area, then it means it is holding a queue of _____ capacity.

- Zero
- **Unbounded (Page 45)**
- Defined
- Bounded

Unbounded Capacity: The queue has infinite length; thus the sender never blocks.

167. _____ are lowest prioritized processes for scheduling in Kernel Group.

- **User processes (Page 90)**
- Character I/O device control processes
- Block I/O control processes
- File manipulation processes

example. In decreasing order of priority, the kernel bands are:

- Swapper
- Block I/O device control processes
- File manipulation
- Character I/O device control processes
- User processes

168. In round-robin (RR) scheduling algorithm, in case the time quantum is very large (infinite), the RR policy remains the same as the _____ policy.

- LIFO (Last in First Out)
- **FCFS (First Come First Served) (Page 86)**
- SJF(Shorted Job First)
- SRTF (Shortest Remaining Time First

The performance of RR algorithm depends heavily on the size of the time quantum. If the time quantum is very large (infinite), the RR policy remains the same as the FCFS policy. If the time quantum is very small, the RR approach is called the **processor**

169. Consider a scenario in which one process P1 enters in its critical section, no other process is allowed to execute in its critical section. This is called _____

- **Mutual exclusion (page 98)**
- Context switching
- Multithreading
- Progress

1. Mutual Exclusion
If process P_i is executing in its critical section, then no other process can be executing in their critical section.

170. Multilevel feedback _____ scheduling allows a process to move between queues.

- List
- **Queue (page 89)**
- Array
- Time

Multilevel Feedback Queue Scheduling
Multilevel feedback queue scheduling allows a process to move between queues. The

171. The link between two processes P and Q to send and receive messages is called _____

- **Communication link (page 43)**
- Message-Passing link
- Synchronization link
- All of the mentioned

size. If processes P and Q want to communicate, a communication link must exist between them and they must send messages to and receive messages from each other through this link. Here are several methods for logically implementing a link and the send

172. _____ command in Linux helps to create a new directory

- ls
- cp
- mv
- **mkdir (page 26)**

Creating Directories
You can use the `mkdir` command to create a directory.

173. _____ is the basis of queuing theory which is branch of mathematics used to analyze systems involving queues and servers.

- **Little's Formula (page 93)**
- Deterministic Modeling
- Queuing Theory
- Queuing Analysis

This formula is called the **Little's formula**, which is the basis of **queuing theory**, a branch of mathematics used to analyze systems involving queues and servers.

174. TSL based solution can implement mutual exclusion, it abbreviates _____

- **TestAndSet Instruction (page 106)**
- Test Swap Instruction
- Trap Set Solution
- Test Set Instrument

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If the machine supports TestAndSet instruction, then we can implement mutual exclusion by declaring a Boolean variable lock, initialized to false. The structure of process P becomes:

175. You can have a thread wait for another thread within the same process by using the _____system call

- **Pthread_join() (page 73)**
- Pthread_exit()
- Pthread_create()
- Pthread_terminate()

Joining a Thread
You can have a thread wait for another thread within the same process by using the pthread_join() call. Here is the syntax of this call.

176. A signal is an event generated to get attention of

- Hard disk
- **Process (page 7)**
- Trap
- Interrupts

A **signal** is an event generated to get attention of a process.

177. Both in two atomic instructions TestAndSet and swap two critical section requirements are satisfied but not _____

- Mutual exclusion
- Progress
- **Bounded waiting (page 106)**
- Assumptions

Just like the TSL-based solution shown in this section, the above Swap-based solution is not good because even though mutual exclusion and progress are satisfied, bounded waiting is not. In the next lecture, we will discuss a good solution for the critical section

178. You can use the rm file 1 command to _____file 1.

- Retrieve
- **Remove (page 27)**
- Make
- Modify

Removing Files
You can use the `rm` command to remove files. You can use the `rm file1` command to remove file1. You can use the first command the following command

179. Shared libreries and kernel modules are stored in _____directory.

- /bin
- /dev
- /boot
- /lib (page 23)
- **/lib** Shared libraries and kernel modules are stored in this directory. The

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180. Use of the semaphore create a problem of busy waiting, this wastes CPU cycle that some other process may be able to use productively. This type of semaphore is also called_____

- Semaphore S
- **Spinlock (page 109)**
- Locking Semaphore
- Mutex

shared among many processes. Busy waiting wastes CPU cycles that some other process may be able to use productively. This type of semaphore is also called a **spinlock** (because the process spins while waiting for the lock). Spinlocks are useful in

181. As a result of <Ctrl-C>, a SIGINT signal is sent to a process. Signal number for SIGINT is _____

- 15
- 1
- 9
- **2 (page 66)**

Terminating a process

You can terminate a foreground process by pressing `<Ctrl-C>`. Recall that this key press sends the SIGINT signal to the process and the default action is termination of the process. Of course, if your foreground process intercepts SIGINT and ignores it, you

specified signal is to be sent. For example, `kill -2 1234` command sends signal number 2 (which is also called SIGINT) to the process with ID 1234. The default actior

182. The section of code after the critical section is called _____.

- Crystal section
- Entry section
- **Remainder section (page 98)**
- Exit section

called the **entry section**. The remaining code is the **remainder section**. T

183. To prevent a process from keeping CPU for too long, control is shifted to another ready process by invoking a timer _____.

- When running process finishes its task
- Every time a free space is available in main memory
- **Every time a process finishes its time slice (page 11)**
- When running process makes a system call

process. Thus we can use the timer to prevent a program from running too long. In the most straight forward case, the timer could be set to interrupt every N milliseconds, where N is the **time slice** that each process is allowed to execute before the next process gets control of the CPU. The OS is invoked at the end of each time slice to perform

184. ---------- View is that operating system is resource manager.

- Top-down
- Single-up
- Down-up
- **bottom-up (page 3)**

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computer system efficiently, fairly, and securely. So, the bottom-up view is that operating system is a resource manager who manages the hardware and software resources in the computer system.

185. Co-operating process sharing a piece of code are executed periodically to solve _____

- Waiting problem
- Non-concurrency problem
- Aging problem
- **Critical section problem (page 97)**

The Critical Section Problem

Critical Section: A piece of code in a cooperating process in which the process may updates shared data (variable, file, database, etc.).

Critical Section Problem: Serialize executions of critical sections in cooperating processes.

186. Software section to critical section problem can run only in environment _____

- Multiprocessor
- Multithreading
- **Uniprocessor (Page 105)**
- Separate address spacing

The critical section problem can be solved simply in a uniprocessor environment if

187. _____ command displays the contents of current working directory.

- **Is (Page 25)**
- cs
- mv
- cp

You can display the contents (names of files and directories) of a directory with the `ls` command. Without an argument, it assumes your current working directory. So,

188. When processes communicate with each other, they perform communication through synchronization and utilizing separate address spaces. This action is termed as _____

- Process management
- Synchronization
- **Inter Process Communication (Page 43)**
- Direct/indirect Communication

Interprocess Communication (IPC)

IPC provides a mechanism to allow processes to communicate and to synchronize their actions without sharing the same address space. We discuss in this section the various

189. We can use semaphore to deal with the number of _____ process critical section problem.

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- 1
- **n (Page 108)**
- n-1
- 2n

We can use semaphores to deal with the n-process critical section problem. The n

190. Which is not considered a role of an operating system?

- Memory allocation and de-allocation of a process
- **Synchronization of cooperating processes (Page 2+3)**
- Program execution
- Protection of a user's file and directories

The above issues require the operating system to provide the following services and much more:

- Synchronize executions of cooperating processes

191. In system call that creates a pipe for IPC. If one of the operation from read/write would fail then what will be returned as result?

- -1+1
- **-1 (Page 46)**
- 1
- 0

Each array element stores a file descriptor. fd[0] is the file descriptor for the read end of the pipe (i.e., the descriptor to be used with the read system call), whereas fd[1] is the file descriptor for the write end of the pipe. (i.e., the descriptor to be used with the write system call).The function returns -1 if the call fails. A pipe is a bounded buffer and the

192. Which of the following is not true for a time-sharing system?

- Multi user
- Iterative
- **Rigid time requirements (Page 06 + 77)**
- CPU sit idle when a process requires I/O operation

A **time-sharing system** is multi-user, multi-process, and interactive system. This means that it allows multiple users to use the computer simultaneously. A user can run one or

In multiprogramming, a process is executed until it must wait, typically for the completion of some I/O request. In a simple computer system, the CPU would then sit idle; all this waiting time is wasted. Multiprogramming entails productive usage of this

193. Which could not be the advantage of thread?

- **Separate address space (Page 69)**
- Quick response
- Economical
- Best in multiprocessing environment

process to which they belong. Code sharing allows an application to have several different threads of activity all within the same address space.

194. OS helps manages the following except

- Application software
- Memory
- Virtual memory
- **Bus speed of the system (Page 1+3+Google)**

Operating system, which manages the use of the hardware among the various application programs for the various users and provides the user a relatively

A computer system has many hardware and software resources that may be required to solve a problem: CPU time, memory space, file storage space, I/O devices etc. The operating system acts as the manager of these resources, facing numerous and possibly

The **operating system manages virtual** address spaces and the assignment of real **memory** to **virtual memory**. Address translation

195. In operating system, _____ command is used to copy files to same location or different location.

- Copy
- Copi
- Cp
- **Both Copy and CP (Page 3+27)**

the difficult work of interacting with computer hardware. Thus, you can use a command like `copy file1 file2` to copy 'file1' to 'file2' and let the operating system

Copying Files

You can use the `cp` command for copying files. You can use the `cp file1 file2` command to copy file1 to file2. The following command can be used to

196. _____ scheduling algorithm can be preemptive or non-preemptive.

- First Come First Serve

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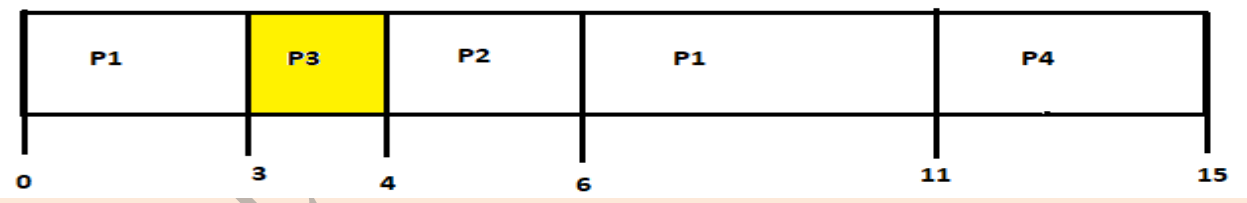
- **Shortest Job First (Page 82)**
- Round Robin
- Priority

The SJF algorithm may either be preemptive or non-preemptive. The choice arises

197. The main characteristic of real time system is.
- Efficiency
- Large virtual memory
- Large secondary storage device
- **Usability (Google)**

product. The properties of six of the most important nonfunctional requirements for real-time systems are analyzed: timing, reliability, safety, security, usability, and maintainability. For each type of requirement, we

198. Processes P1,P2,P3,P4 enter the ready queue at time 0,3,3,11 with burst times P1=8, P2=2, P3=1, P4=1. With shortest remaining time first ,_____ process is given CPU at time=3.
- P1
- P2
- **P3 Conceptual**
- P4



199. If size of a process is 376052 bytes and its smallest physical memory address is 242785.Its address space cannot exceed beyond_____.
- 376053
- 376051
- 618836
- **618837 (Conceptual Page 11)**

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In Figure 3.2, we use an example to illustrate how the concept outlined above works. The base and limit registers are initialized to define the address space of a process. The process starts at memory location 300040 and its size is 120900 bytes (assuming that memory is byte addressable). During the execution of this process, the CPU insures (by using the logic outlined in Figure 3.1) that all the addresses generated by this process are greater than or equal to 300040 and less than (300040+120900), thereby preventing this

200. Which of the following is a correct command line for running in background?

- cat code.c | grep print %
- **cat code.c | grep print & (Page 62)**
- cat code.c | grep print #
- cat code.c | grep print @

used for communication between the two processes. Note that the server process is run in the background by terminating its command line with an ampersand (&).

201. Which of the following command is used to read data from file named academics?

- con<academics
- socket>academics
- ran>academics
- **cat<academics (Page 55)**

redirection. In these examples, the `cat` and `grep` commands read input from the Phones file and not from keyboard.

```
$ cat < Phones
```

202. Using hardware solution to synchronization for complex problems, introduce a new synchronization tool known as_____.

- Trap
- **Semaphore (Page 108)**
- Swap
- TestAndSet

Hardware solutions to synchronization problems are not easy to generalize to more complex problems. To overcome this difficulty we can use a synchronization tool called a semaphore. A **semaphore** S is an integer variable that, apart from initialization is accessible only through two standard atomic operations: wait and signal. These

203. In LINUX directory structure, there is_____ root directory.

- **1 (Page 23)**
- 2

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- 4
- 3

to a drive letter in Windows (C:\, D:\, etc.) except that in the Linux directory structure there is only one root directory and everything falls under it (including other file systems and partitions). The root directory is

204. The time interval from submission of a process to the time when its output is generated is termed as_____.

- **Turnaround time (Page 80)**
- CPU burst
- Responsive time
- Throughput

Turnaround time: The interval from the time of submission to the time of completion is the **turnaround time.** Turnaround time is the sum of the periods spent

205. The correct command for compiling C program named program.c in LINUX environment is

- gcc -z FirstProgram program.c
- gcc program.c -m FirstProgram
- gcc -o FirstProgram program.c
- **gcc program.c -o First Program (Page 28)**

```
$ gcc program.c -o assignment
```

206. The Kernel is _____ user threads.

- is part of
- **unaware of (Page 70)**
- aware of
- the creator of

management with no support from the kernel. Since the kernel is unaware of user-level threads, all thread creation and scheduling are done in the user space without the need for

207. _____ scheduling algorithm is sometimes called shortest remaining time first scheduling algorithm.

- Priority scheduling
- Non-preemptive SJF
- **Preemptive Shortest Job First (Page 82)**
- FCFS

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finish its CPU burst. Preemptive SJF scheduling is sometimes called **shortest-remaining-time-first** scheduling.

208. You can use the bg command to put the current or a suspended process into the background. What is the correct syntax of bg command from the following?

- bg [%id]
- bg [job#]
- bg [id#]
- **bg [%job_id] (Page 65)**

You can use the bg command to put the current or a suspended process into the background. Here is the syntax of the command.

```
bg [%job_id]
```

209. The region in the memory that a process is allowed to access is known as_____.

- **Address space (Page 10)**
- Instruction
- Operation
- Register

The region in the memory that a process is allowed to access is known as **process address space**. To ensure correct operation of a computer system, we need to ensure that

210. In Unix/Linux by default the standard output file is attached to the

- File
- **Screen (Page 52)**
- Printer
- Scanner

  - Standard output: 1 (display screen)

211. The procedure "The time at which the process finished working MINUS the arrival time of the process MINUS CPU burst for that process" will help calculate the _____.

- Non-preemptive Shortest Job First scheduling.
- **Preemptive Shortest Job First scheduling. (Page 82**
- FCFS
- RR Scheduling

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Non-Preemptive = Waiting time = Finish Time – Arrival Time
Preemptive = Waiting time = Finish Time – Arrival Time – Burst time

212. Processes P1, P2, P3 having burst time P1=3, P2=4, P3=3 and turnaround times P1=7, P2=10, P3=9 enter the ready queue at Time=0. With RR scheduling waiting time for process P2 is_____.

- 4
- 0
- 3
- **6 Conceptual**

Process	Burst Time	Turnaround time	Waiting time
P1	3	7	7-3= 4
P2	**4**	**10**	**10-4 =6**
P3	3	9	9-3= 6

213. _____ Scheduler takes the process from the ready queue and assigns the CPU with the help of Dispatcher.

- Swapper
- **Short term (Page 79)**
- Long term
- Medium term

in the ready queue to be executed. The **short-term scheduler** (i.e., the CPU scheduler) selects a process to give it the CPU. It selects from among the processes in memory that are ready to execute, and invokes the dispatcher to have the CPU allocated to the selected process.

214. To display all processes _____ option is used with ps command/

- -l
- -a
- **-e (Page 63)**
- -u

`ps` gives a snapshot of the current processes. Without options, ps prints information about processes owned by the user. Some of the commonly used options are `-u`, `-e`, and `-l`.

- `-e` selects all processes

215. Clock interrupt handler is an example of kernel level thread that is not executed for serving a request by a user thread. It is named as_____.

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- EINVAL
- EAGAIN
- **LWP (Page 72)**
- POSIX

kernel's knowledge. One kernel thread is assigned per LWP. Some kernel threads have no LWP associated with them because these threads are not executed for servicing a request by a user-level thread. Examples of such kernel threads are clock interrupt handler, swapper, and short-term (CPU) shceduler.

216. Consider three processes in scheduling. Here are the waiting times for three process P1=4,P2=2,P3=6.Which one of the following is correct average waiting time.

- 2
- **4 Conceptual**
- 3
- 6

Here are the waiting times for the three processes and the average waiting time per process.

Waiting times **P1 = 4; P2 = 2 ; P3 = 6;**

Average waiting time: **(4+2+6)/3 = 4**

Average waiting time= 4

217. Processes P1,P2,P3 having burst time P1=3,P2=4,P3=3 and turnaround time P1=7,P2=10,P3=9 enter the ready queue at T=0 waiting time for process P2 is_____.

- 3
- 4
- **6 Conceptual**
- 0

Waiting Time = Turnaround Time – Burst Time

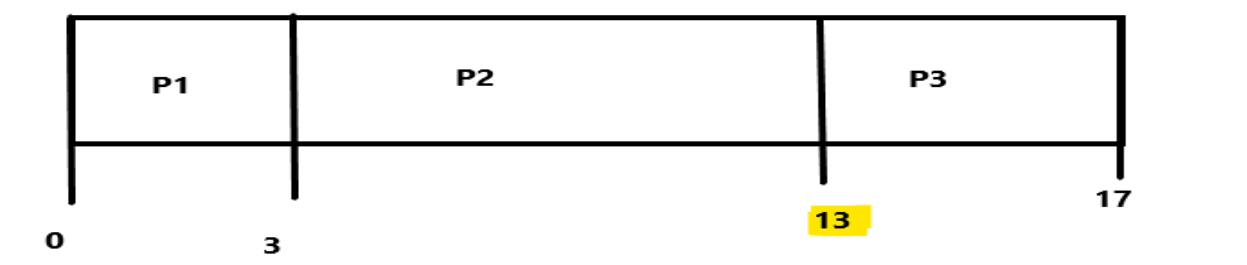
Process Id	Burst Time	Turnaround Time	Waiting Time
P1	3	7	7-3 = **4**
P2	4	10	10-4 = **6**
P3	3	9	9-3 = **6**

218. P1,P2,P3,P4,P5 are five processes with tick numbers P1=1,P2=2,P3=4,P4=2,P5=4 then considering Lamport's bakery algorithm _____process enters critical section after P1.

- **P2 (Not Sure)**
- P3
- P4
- P5

219. Four processes P1, P2, P3, P4 enter the ready queue at time 0 with burst times P1=3, P2=10, P3=4. Waiting time of P3 would be _____.

- 12
- **13 (Conceptual not sure)**
- 14
- 17



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